Project Overview

This project uses a **NodeMCU (ESP8266)** to:

- Automatically turn street lights ON/OFF based on light intensity
- Monitor voltage and current consumption
- Detect light failure
- Send real-time data to a web dashboard
- Allow remote manual control

It reduces energy consumption and enables smart city integration.

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2. Objectives

- Reduce electricity wastage
- Automate street light operation
- Enable remote monitoring
- Detect faulty lights
- Build low-cost IoT-based infrastructure

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3. Components Required

Component	Quantity
NodeMCU ESP8266	1
LDR (Light Dependent Resistor)	1
10k Resistor	1
Relay Module (5V)	1
ACS712 Current Sensor	1
Street Light (LED Bulb)	1

Component	Quantity
Power Supply	1
Jumper Wires	As needed

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4. Connections:

- **LDR**
  - One end → 3.3V
  - Other end → A0 (via voltage divider)
- **Relay Module**
  - IN → D1
  - VCC → 5V
  - GND → GND
- **ACS712**
  - VCC → 5V
  - GND → GND
  - OUT → A0 (using analog multiplexer or separate ADC setup)

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5. Software Requirements

- Arduino IDE
- ESP8266 Board Package
- Libraries:
  - ESP8266WiFi.h
  - ESP8266WebServer.h

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6. Working Principle

1. LDR senses sunlight.
2. If light intensity is low → Street light turns ON.
3. If light intensity is high → Street light turns OFF.
4. ACS712 measures current flow.
5. If no current detected → Fault alert triggered.
6. NodeMCU hosts a local web server.
7. User can monitor:
    - Light status
    - Current usage
    - Manual ON/OFF control